

Ambient WiFi as a Cheap Radar for Automotive SLAM: Centimetre Localization, a Phantom Ceiling on Mapping, and What That Ceiling Is — in Simulation and on \$30 of Silicon

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Abstract—We ask whether ambient sub-7 GHz WiFi, received on a moving vehicle and processed with commodity channel state information (CSI), can serve as a low-cost radar/LiDAR replacement for simultaneous localization and mapping (SLAM). Using a physics-based, ray-traced (Sionna RT) substrate and, for the first time in this line of work, a \$30 ESP32 hardware testbed, we find a consistent split. *Localization* is centimetre-level and, we show, structurally robust: because the estimator is odometry-anchored and the bistatic CSI measurements are refinement-only, absolute trajectory error is insensitive to access-point-position error up to tens of metres and to sudden, total measurement blackouts. *Mapping*, by contrast, hits a hard ceiling: with realistic commodity CSI, most detections are phantoms ($\approx 89\%$) accompanied by a several-metre range bias — a front-end and geometry limit, not a path-discrimination or resolution limit, and one a learned filter cannot repair. Comparing the same substrate against real LiDAR (KITTI) and a simulated 77 GHz FMCW radar shows the ceiling is neither carrier-specific nor WiFi-specific: the radio carrier does not matter, geometry does (a monostatic on-vehicle transmitter nearly eliminates the phantoms), and wider bandwidth makes them worse. Finally, we reproduce the delay-domain reading on real ESP32 silicon. The practical message: WiFi localization rivals far costlier sensors, while WiFi mapping is capped by a front-end/geometry phantom ceiling that we characterize and, in part, show how to escape.

Index Terms—WiFi sensing, channel state information, SLAM, passive radar, integrated sensing and communication, automotive, ESP32, low-cost hardware, multipath.

I. INTRODUCTION

THE exteroceptive sensors that anchor vehicle SLAM — LiDAR and radar — are also the costly part of the perception stack. A research-grade spinning LiDAR runs into the thousands of dollars and an automotive radar into the tens to hundreds, while the WiFi silicon a vehicle already carries costs tens. If ambient WiFi — signals radiated anyway, by infrastructure that already exists, received on an antenna already on the vehicle — could stand in for those sensors, the sensing bill for autonomy would change by orders of

magnitude. This paper asks the question without hedging: *can ambient sub-7 GHz WiFi replace radar or LiDAR as a SLAM front-end?*

The honest answer is neither “yes” nor “no” but a boundary, and the boundary is the contribution. WiFi replaces the costly sensors for one half of SLAM — localization — and fails at the other — mapping — and the failure is not where the literature, including our own earlier feasibility study [1], assumed it was. We locate that failure precisely, show it is *not specific to WiFi*, and then reproduce the underlying reading on real hardware.

Localization is centimetre-level and, we show, *structurally* robust: because the estimator is anchored by vehicle odometry and the bistatic CSI measurements are refinement-only, absolute trajectory error is insensitive to access-point (AP) position error up to tens of metres and to sudden, total measurement blackouts. *Mapping* hits a hard ceiling: with realistic commodity CSI, the overwhelming majority of detections ($\approx 89\%$) are *phantoms* — they match no real propagation path — accompanied by a several-metre range bias. This is a front-end and geometry limit, not a path-discrimination or resolution limit, and a learned filter cannot repair it, because one cannot select real paths from a set in which most detections are not real paths.

To test whether this ceiling is a property of WiFi or of RF sensing more broadly, we run the *same* substrate against real LiDAR (KITTI [2]) and a simulated 77 GHz FMCW radar. The result is decisive: the radio carrier does not matter, geometry does — a *monostatic*, on-vehicle transmitter nearly eliminates the phantoms — and wider bandwidth makes them *worse*. Finally, licensed by that carrier-independence, we move the claim off simulation entirely and capture, parse, and delay-profile real HT40 CSI on two \$30 ESP32-S3 boards.

Contributions

- A physics-based feasibility study of on-vehicle, mobile, outdoor WiFi as a SLAM front-end — an unoccupied point in the design space (prior passive-WiFi-

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radar is indoor/static/infrastructure-fixed; the one moving-platform passive radar uses 5G).

- Identification and mechanism of the *mapping phantom ceiling* ($\approx 89\%$ phantom detections + range bias), and proof it is not a discrimination or resolution limit and not repairable by a learned filter.
- Evidence the ceiling is *universal*: carrier-independent and not WiFi-specific, via a controlled comparison against LiDAR and a 77 GHz radar on one substrate; geometry (monostatic vs bistatic), not carrier or bandwidth, is decisive.
- A robustness characterization: localization is structurally tolerant of access-point–position error and measurement blackouts (Sec. V-B).
- The first delay-resolved reflector reading from an ESP32 on \$30 of hardware, moving the claim off simulation (Sec. VI).

II. RELATED WORK

WiFi/CSI sensing is mature but overwhelmingly *indoor, static, and infrastructure-fixed*. Through-the-wall and device-free sensing [3], [4], [5], CSI-based ranging and imaging [6], [7], [8], and pose or point-cloud recovery [9], [10] all assume fixed transmitters and a static or slowly-varying scene. None places the sensing platform on a moving vehicle outdoors.

WiFi-based SLAM takes two forms. Channel- and multipath-based SLAM treats reflectors as virtual anchors [11], [12], [13], [14], and radio-fingerprint or P2SLAM approaches localize against known access points [15], [16] — both *infrastructure-bound*: they need the transmitters’ positions and are demonstrated indoors. The alternative uses WiFi only to *assist* a primary ranging sensor (LiDAR or IMU) [17], [18], [19], not to replace it. No prior work evaluates WiFi as a stand-alone, on-vehicle, outdoor *replacement* front-end.

Integrated sensing and communication (ISAC/JCAS) envisions the same waveform serving communication and sensing [20], [21], [22], [23], but its SLAM instantiations are early [24]. At mmWave, 60 GHz 802.11ad has been repurposed as a radar [25], [26]; the only *moving-platform* passive radar we are aware of uses 5G, not WiFi [27].

The gap. On-vehicle, mobile, outdoor WiFi (802.11) as a radar/LiDAR replacement for SLAM is an unoccupied point in this design space. This paper occupies it, and — unlike the infrastructure-bound line above — shows that the strongest configuration (monostatic, on-vehicle transmitter) needs *no* known AP positions at all.

III. METHOD: ONE SUBSTRATE, THREE SENSORS, ONE TESTBED

A single substrate — identical scenes, trajectories, footprint ground truth, and metric suite — carries WiFi, LiDAR, and 77 GHz radar, so differences are attributable to the sensor, not the harness. The same detection chain then runs on real ESP32 hardware.

A. Bistatic geometry

A vehicle carrying a receive antenna array drives through an outdoor scene illuminated by N_{AP} ambient WiFi access points at known positions. Each specular component follows an access point $\rightarrow$ reflector $\rightarrow$ vehicle path, so the measured path length $\rho = \|\mathbf{p}_{\text{AP}} - \mathbf{r}\| + \|\mathbf{r} - \mathbf{p}_{\text{veh}}\|$ places the reflector $\mathbf{r}$ on an ellipse with foci at the AP and the vehicle. Given the arrival azimuth ϕ , the reflector range s along $\hat{\mathbf{u}}(\phi)$ solves

$$s = \frac{\rho^2 - d_{\text{AP}}^2}{2(\rho - \mathbf{d}_{\text{AP}} \cdot \hat{\mathbf{u}}(\phi))}, \quad \mathbf{d}_{\text{AP}} = \mathbf{p}_{\text{AP}} - \mathbf{p}_{\text{veh}}, \quad (1)$$

with $d_{\text{AP}} = \|\mathbf{d}_{\text{AP}}\|$; direct-path ($s \leq 0$) and implausible solves are rejected. A *monostatic* variant places the transmitter on the vehicle itself ($\mathbf{p}_{\text{AP}} \rightarrow \mathbf{p}_{\text{veh}}$): the measurement becomes a round-trip range, and (1) needs no external AP position at all — a fact that turns out to decide both the phantom rate (Sec. V) and the AP-error robustness (Sec. V-B).

B. Ray-traced channel and scenes

Channels are generated with Sionna RT 2.0 (Mitsuba/Dr.Jit), which ray-traces the scene to produce per-path delays, angles, and interaction types and the frequency-domain channel across N_{sc} subcarriers and N_{rx} antennas per vehicle pose; additive white Gaussian noise is applied at a configured SNR. Radar is electromagnetic, so the same tracer models it natively; LiDAR requires an optical (diffuse-scattering) treatment (Sec. III-E).

C. CSI front-end: joint 2-D (delay–angle) MUSIC

We compare two sensing tiers. The *oracle* front-end reads Sionna’s true single-scatter path parameters, isolating the geometry from estimation error. The *realistic* front-end estimates delay and azimuth from noisy CSI with MUSIC. A key refinement is *joint 2-D (delay–angle) MUSIC* with 2-D spatial smoothing (SVD subspace; delay grid bounded to the unambiguous $1/\Delta f$ range), which recovers each path’s delay and angle *together*, so association is intrinsic rather than paired by sorted index. Array-relative electrical angles map to world azimuth by $\sin \beta = -\sin \theta$ for the lateral ULA.

D. Bistatic-ellipse mapping and particle-filter SLAM

A Rao–Blackwellized particle filter propagates the vehicle pose from odometry and weights particles by the bistatic consistency (1) of each detection; triangulated reflectors are accumulated and consensus-filtered into a landmark map. We report trajectory error (ATE, RPE) and, for the map, symmetric Chamfer distance, occupancy IoU, and *directional* accuracy and completeness (a passive-WiFi map illuminates only a subset of surfaces). Map ground truth is each scene object’s facade footprint.

E. Complexity and real-time feasibility

The 2-D MUSIC front-end is the cost driver: per AP per frame it forms a spatially-smoothed covariance over the delay $\times$ angle grid and takes its eigendecomposition, an $\mathcal{O}(N^3)$ operation in the smoothed dimension N . Two facts keep this

TABLE I

THE MAPPING CEILING DECOMPOSES INTO THREE MECHANISMS; THE SMALLEST IS THE ONE PAPER I NAMED. (CONTROLLED / STREET)

Mechanism	Magnitude	Role
Phantom detections	89.2%/89.5%	dominant (no real path)
Estimation range bias	6.45 m (controlled)	ruins correct paths
Path discrimination	2–8%	real, but <i>smallest</i>

practical. First, N is small (a bounded delay grid and a single ULA), and the work is per-AP-per-frame and embarrassingly parallel. Second — and more consequentially — Sec. V shows that MUSIC is precisely what *creates* the mapping phantoms, and that a calibrated constant-false-alarm-rate (CFAR) detector in the monostatic geometry both removes most phantoms *and* is far cheaper: the monostatic delay-domain chain we port to hardware costs ≈ 194 kFlop and ≈ 26 KB per sweep, which an ESP32-S3 executes with $\approx 250\times$ headroom. Real-time on-vehicle execution and the mapping fix therefore point the same way.

F. LiDAR and 77 GHz radar baselines

The LiDAR baseline is an Ouster-class spinning sensor, ray-traced with a diffuse-scattering model and a scan-to-map ICP back-end that is *externally validated* on real KITTI sequences ($\approx 0.3\%$ drift), so its simulated numbers are anchored to reality. The radar baseline is a 77 GHz FMCW sensor with a range–azimuth CFAR chain. One honest caveat propagates from the radar study: our point-to-point ICP back-end cannot recover *rotation* from spinning-radar point clouds (the yaw cost is flat), so the radar/WiFi mapping ablation (Table II) is scored under *ground-truth poses* — it isolates the front-end and geometry, which is exactly the quantity of interest.

G. ESP32-S3 hardware testbed

Two \$30 ESP32-S3 boards realise the monostatic geometry directly: one illuminates (HT40, 40 MHz) and one logs CSI ≈ 0.5 m away, so the round-trip delay taps *are* echo ranges. Only the *excess* delay (relative to the line-of-sight arrival) is needed, and that is preserved on commodity CSI even though absolute time-of-flight is not; the 0.5 m baseline is under 7% of one 40 MHz range cell, so the rig is monostatic to within measurement. Using 2.4 GHz hardware is licensed by the carrier-independence result (Table II, B→C). Details in Sec. VI.

TABLE II

ABLATION ON ONE SUBSTRATE (CONTROLLED_WALL, 3 SEEDS). GEOMETRY, NOT CARRIER OR BANDWIDTH, SETS THE PHANTOM RATE.

Cell	Front-end / geometry	Carrier/BW	Phantom rate
M	MUSIC, bistatic	5.2 GHz/160 MHz	— (IoU 0.003)
A	CFAR, bistatic	5.2 GHz/160 MHz	$18.2 \pm 0.6\%$
B	CFAR, monostatic	5.2 GHz/160 MHz	$0.1 \pm 0.2\%$
C	CFAR, monostatic	77 GHz/160 MHz	0.0%
D	CFAR, monostatic	77 GHz/4 GHz	$9.0 \pm 1.5\%$

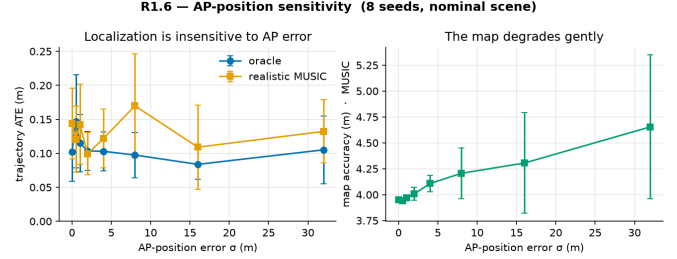


Fig. 1. [R1.6] Trajectory ATE is insensitive to AP-position error to 32 m (structural: odometry-anchored, refinement-only measurements); the map degrades gently and monotonically.

IV. FEASIBILITY: LOCALIZATION WORKS, MAPPING IS TWO-TIER

V. THE MAPPING CEILING AND ITS MECHANISM

A. Universality: WiFi vs 77 GHz radar

B. Robustness to AP-position error and measurement loss

VI. HARDWARE VALIDATION ON \$30 OF SILICON

VII. DISCUSSION

VIII. CONCLUSION

REPRODUCIBILITY

All code, configs, and result data are in the repository above (AGPL-3.0); the simulation is Sionna RT; the hardware pipeline and firmware are included.

REFERENCES

- [1] M. Fetna, “Ambient WiFi as a radar replacement for automotive SLAM: A physics-based feasibility study,” 2026, submitted. Artifacts: <https://doi.org/10.5281/zenodo.21247288>.
- [2] A. Geiger, P. Lenz, and R. Urtasun, “Are we ready for autonomous driving? the KITTI vision benchmark suite,” in *IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2012.
- [3] K. Chetty, G. E. Smith, and K. Woodbridge, “Through-the-wall sensing of personnel using passive bistatic WiFi radar at standoff distances,” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 50, no. 4, pp. 1218–1226, 2012. [Online]. Available: <https://ieeexplore.ieee.org/document/6020778>
- [4] M. Zhao, T. Li, M. Abu Alsheikh, Y. Tian, H. Zhao, A. Torralba, and D. Katabi, “Through-wall human pose estimation using radio signals,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018. [Online]. Available: <https://rfpose.csail.mit.edu/>
- [5] M. Zhao, Y. Tian, H. Zhao, M. Abu Alsheikh, T. Li, R. Hristov, Z. Kabelac, D. Katabi, and A. Torralba, “RF-based 3D skeletons,” in *Proceedings of the ACM SIGCOMM Conference*, 2018.
- [6] W. Li, R. J. Piechocki, K. Woodbridge, C. Tang, and K. Chetty, “Passive WiFi radar for human sensing using a stand-alone access point,” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 59, no. 3, pp. 1986–1998, 2021. [Online]. Available: <https://discovery.ucl.ac.uk/id/eprint/10103371>

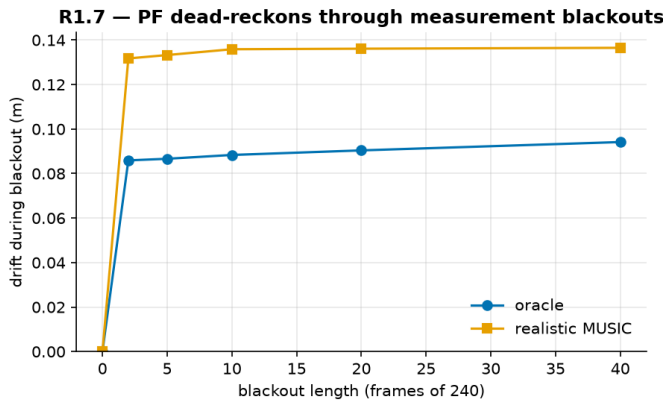


Fig. 2. [R1.7] The particle filter dead-reckons through a total measurement blackout: a 40-frame (17% of run) outage adds ≈ 0.1 m drift and re-locks immediately.

[7] Wang, Zhang, Wu, Chen, Xu, and Guo, “Bistatic passive sensing via CSI power,” arXiv preprint arXiv:2511.22144, 2025. [Online]. Available: <https://arxiv.org/abs/2511.22144>

[8] T. Määtä, S. Sharifpour, M. Bordallo López, and C. Álvarez Casado, “Spatio-temporal 3D point clouds from WiFi-CSI data via transformer networks,” 2024, arXiv:2410.16303.

[9] J. Geng, D. Huang, and F. De la Torre, “DensePose from WiFi,” 2022, arXiv:2301.00250.

[10] Y. Yan *et al.*, “Person-in-WiFi 3D: End-to-end multi-person 3D pose estimation with Wi-Fi,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2024/papers/Yan_Person-in-WiFi_3D_End-to-End_Multi-Person_3D_Pose_Estimation_with_Wi-Fi_CVPR_2024_paper.pdf

[11] C. Gentner, T. Jost, W. Wang, S. Zhang, A. Dammann, and U.-C. Fiebig, “Multipath assisted positioning with simultaneous localization and mapping,” *IEEE Transactions on Wireless Communications*, vol. 15, no. 9, pp. 6104–6117, 2016.

[12] M. Ulmschneider, C. Gentner *et al.*, “Multipath assisted positioning for pedestrians using LTE signals,” *Mobile Information Systems*, vol. 2017, p. 9170746, 2017. [Online]. Available: <https://www.hindawi.com/journals/misy/2017/9170746/>

[13] E. Leitinger, L. Wielandner, A. Venus, and K. Witrals, “Multipath-based SLAM with cooperation and map fusion in MIMO systems,” arXiv preprint arXiv:2405.02126, 2024. [Online]. Available: <https://arxiv.org/abs/2405.02126>

[14] X. Li, X. Cai, E. Leitinger, and F. Tufvesson, “A belief propagation algorithm for multipath-based SLAM with multiple map features: A mmWave MIMO application,” arXiv preprint arXiv:2403.10095, 2024. [Online]. Available: <https://arxiv.org/abs/2403.10095>

[15] A. Arun, R. Ayyalasomayajula, W. Hunter, and D. Bharadia, “P2SLAM: Bearing based WiFi SLAM for indoor robots,” *IEEE Robotics and Automation Letters*, vol. 7, no. 2, 2022.

[16] R. Liu, B. P. L. Lau, K. Ismail, A. Chathuranga, C. Yuen, S. X. Yang, Y. L. Guan, S. Mao, and U.-X. Tan, “Exploiting radio fingerprints for simultaneous localization and mapping,” *IEEE Pervasive Computing*, 2023, arXiv:2305.13635.

[17] A. Arun, W. Hunter, R. Ayyalasomayajula, and D. Bharadia, “Vi-WiD: Leveraging WiFi for robust and resource-efficient SLAM,” 2022, arXiv:2209.08091.

[18] K. Ismail, R. Liu, Z. Qin, A. Athukorala, B. P. L. Lau, M. Shalihan, C. Yuen, and U.-X. Tan, “Efficient WiFi LiDAR SLAM for autonomous robots in large environments,” in *IEEE Int. Conf. on Automation Science and Engineering (CASE)*, 2022, arXiv:2206.08733.

[19] Z. Li, Z. Tang, K. S. Kim, S. Li, and J. S. Smith, “EKF-based fusion of Wi-Fi/LiDAR/IMU for indoor localization and navigation,” 2025, arXiv:2509.23118.

[20] F. Liu, Y. Cui, C. Masouros, J. Xu, T. X. Han, Y. C. Eldar, and S. Buzzi, “Integrated sensing and communications: Toward dual-functional wireless networks for 6G and beyond,” *IEEE Journal on Selected Areas in Communications*, vol. 40, no. 6, pp. 1728–1767, 2022.

[21] Zhang *et al.*, “Joint communication and sensing (JCAS): a survey,” arXiv preprint arXiv:2102.12780, 2021. [Online]. Available: <https://arxiv.org/abs/2102.12780>

[22] Y. Xiong, F. Liu, Y. Cui, W. Yuan, T. X. Han, and G. Caire, “On the fundamental tradeoff of integrated sensing and communications under Gaussian channels,” *IEEE Transactions on Information Theory*, vol. 69, no. 9, pp. 5723–5751, 2023.

[23] “An overview on IEEE 802.11bf: WLAN sensing,” *IEEE Communications Surveys & Tutorials*, 2024. [Online]. Available: <https://arxiv.org/abs/2310.17661>

[24] Gounis, S. A. Tegos, D. Tyrovolas, P. D. Diamantoulakis, and G. K. Karagiannidis, “When SLAM meets wireless communications: A survey,” arXiv preprint arXiv:2602.06995, 2026. [Online]. Available: <https://arxiv.org/abs/2602.06995>

[25] P. Kumari, J. Choi, N. González-Prelcic, and R. W. Heath, “IEEE 802.11ad-based radar: An approach to joint vehicular communication-radar system,” *IEEE Transactions on Vehicular Technology*, vol. 67, no. 4, pp. 3012–3027, 2018. [Online]. Available: <https://arxiv.org/abs/1702.05833>

[26] Han, J. Choi, and R. W. Heath, “Radar imaging based on IEEE 802.11ad waveform in V2I communications,” arXiv preprint arXiv:2208.02473, 2022. [Online]. Available: <https://arxiv.org/abs/2208.02473>

[27] R. Maksymiuk *et al.*, “5G-based passive radar on a moving platform—detection and imaging,” *IET Radar, Sonar & Navigation*, 2024. [Online]. Available: <https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/rsn2.12559>